



Lean Startup: Why Large Software Companies Should Care

Henry Edison
Faculty of Computer Science
Free University of
Bozen-Bolzano
39100, Bozen-Bolzano, Italy
henry.edison@inf.unibz.it

Xiaofeng Wang
Faculty of Computer Science
Free University of
Bozen-Bolzano
39100, Bozen-Bolzano, Italy
xiaofeng.wang@unibz.it

Pekka Abrahamsson
Department of Computer and
Information Science
Norwegian University of
Science Technology
NO-7491 Trondheim, Norway
pekka.abrahamsson@id.ntnu.no

ABSTRACT

Software industry has been shifting from traditional, plan-based software development to agile software development. Agile seems able to prescribe on how to develop a working software faster, but is still unable to give answer what product should be developed. This paper aims to investigate why large companies should adopt Lean startup to seek for radical innovation. This understanding could be used as a basis to confront disruptive innovation where the problem and the solution are both unknown. A single case study was conducted in a large software company. Eight practitioners with different roles were interviewed. We identified the characteristics of internal startups and the lessons learned on practising Lean Startup inside the large software company.

CCS Concepts

•Social and professional topics → Systems development; •Software and its engineering → Agile software development; *Risk management*;

Keywords

Startup, Lean startup, agile, large software companies, innovation, radical innovation

1. INTRODUCTION

After its introduction in 2001, agile methodology has been adopted in both small and large software companies and development teams. Agile seems able to prescribe on how to develop a working software faster, but is still unable to give answer what product should be developed [6]. Although agile also advocates to build the software iteratively, it only works when the the problem is known to the stakeholders. When it comes to innovation, companies are not only required to deliver a product faster to the market, but also to

develop one that the customers will pay for. It is not only about development any more but overall business.

Most large companies are struggling to seek a way to innovate radically. Through radical innovation, companies are able to transform existing market or create a new one [2]. Such turnover signifies the importance of continuous innovation rather than a happen-stance [13, 20]. Therefore, Peter Drucker [10] suggests that innovation must be done in an organised, purposeful and systematic manner.

Corporate Entrepreneurship (CE) is considered as one form of corporate innovation [16]. CE enables companies to continuously innovate and keep their position in the market [16]. The focus is not only in exploring new product and services but also new business domain which differ from prior routines. According to Sharma and Chrisman [27], CE can be done using three ways: corporate venturing (introducing new business), innovation and strategic renewal (reconfiguration of existing business). For both venturing and strategic renewal, innovation becomes the main cause for change.

Eric Ries introduced Lean startup in 2011 [24] as a new approach of entrepreneurship. Instead of emphasising on business plan, Lean startup advocates to build the product iteratively and deliver to the market for earlier feedback. Gilb et. al. [12] refer Lean startup as a more "extreme" agile approach than XP or Scrum to manage system building processes. Even though Ries argues that large companies can benefit from practising Lean startup [24], there is a lack of scientific evidence to support that claim. In this study, we aim at investigating the following research question: *Why should large companies adopt Lean startup?*

The rest of this paper is composed as follows. In Section 2, a brief explanation about startup, innovation in large companies and the principles of Lean startup are discussed. Section 3 presents the empirical research design. The results of this study are reported in Section 4 and discussed in Section 5. The conclusion and future work are identified and outlined in Section 6.

2. LITERATURE REVIEW

2.1 Software Startups

Eric Ries defines a startup as "a human institution designed to create a new product or service under conditions of extreme uncertainty" [24]. Based on this definition, there are three components of a startup, as follows:

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- **Institution.** A startup is about building an institution. It includes management activities such as recruitment, coordination, culture etc.
- **Innovation.** What makes a startup different with other business is that innovation becomes the main driver for the company success.
- **Extreme uncertainty.** A startup works in an extreme uncertainty situation. The path to future is still unclear, not only about the product but also about the market.

In software startups context, Sutton [28] describes their characteristics, as follows:

- **Youth and immaturity.** The founders of the startups are young and inexperienced. They are also immature in development process and organisation management.
- **Limited resources.** The available resources are used for getting the product out, promoting, and seeking for alliance.
- **Multiple influence.** In the early phase, a new company gets pressures from internal (team's inconsistency) and external (investors, customers, partners, competitors).
- **Dynamic technologies and markets.** To enter an existing market or a new market, a software company often develops technologically innovation product. This requires disruptive technologies.

A recent study by Paternoster et. al. [22] finds that there is no agreement on what constitutes a startup. The study also finds 15 different characteristics of startup (see Table 1).

2.2 Innovation in large companies

The constraint of large companies to innovate is not because of lack of innovative ideas or employees [23]. In technological-entrepreneurs should “get out of the building” to involve the customers since day one [5].

based companies, they limit themselves to work only on the ideas within their scope. This condition is referred to as “technology inertia” [11] because they have invested many resources in the existing technology and market. Moreover, the size and complexity of modern business have made the company to be bureaucratic, which has potential to lower company agility to innovate [26]. The failure to generate radical innovation is also caused by “the incumbent’s curse” [7]. This happens when the incumbents focus too much on their current position in the market and satisfying current customers rather than seeking for new pathways to which the new products might lead.

Entrepreneurial activities within existing firms, or in another term corporate entrepreneurship, have gained more attention from practitioners and academics. CE has been suggested as a tool to facilitate a company’s effort to innovate [16]. Entrepreneurship is defined as “the process of creating value by bringing together a unique of resources to exploit an opportunity” [29]. Peter Drucker [10] suggests that innovation is a specific tool to identify the opportunity. In addition, he also claims that innovation creates resources that will be able to generate economic value. Therefore he suggests that innovation must be done in an organised, purposeful and systematic manner. It is not because of the work

of a genius with brilliant ideas but a team with discipline to find the sources of innovative opportunities.

The study by Stevenson and Jarillo [30] finds that the motivation of companies to be engaged in CE is the exceptional growth. To obtain the growth, companies aggressively search for and exploit new opportunities, although they are limited by the availability of their resources. Companies must take risks and innovate proactively. One way of dealing with these issues is through creating new venture [15, 17, 19, 27]. A new venture can be introduced as a separate division of a corporation (internal venture) or as a newly stand-alone company (external venture) created to develop new products and services and introduce them to the market [3]. This study focuses on the internal venture or internal startup. Compared to external startup, internal startup has advantages and disadvantages, as shown in Table 2.

Table 2: Advantages and disadvantages of internal startup

Advantages	Disadvantages
Access to tangible and intangible resources [18]	Costly (in terms managerial involvement and resource commitment) [9, 25]
Use the existing brand and reputation [23]	Possibility to destroy the existing brand and reputation [27]
Add new offerings in a specific market [23]	Culture change [18]

2.3 Agile and Lean startup

Inspired by lean manufacturing principles from Toyota, Eric Ries introduced a new way of entrepreneurship [24]. Lean startup focuses on the efforts that create value to customers and eliminate waste during the development phase. However, since the customers are often unknown, what customers could perceive as value are also unknown. Therefore, entrepreneurs should “get out of the building” to involve the customers since day one [5].

Lean startup has five principles [24], as follows:

- **Entrepreneurs are everywhere.** Anyone can be an entrepreneur without owning a business, either a student or an employee within a corporation.
- **Entrepreneurship is management.** Startup is not only about product development but business development.
- **Validated learning.** To build a sustainable business, entrepreneurs should run experiments and validate what customers need. Therefore, they reveal current and future business prospects.
- **Build-measure-learn.** This is the fundamental activity of a startup. To perceive customer value, an entrepreneur should start a feedback loop that turns an idea into a product then learn whether to pivot or persevere. This can be done by developing a minimum viable product (MVP) using an agile development method. MVP is used as a tool to collect customer feedback on the product. Then the feedback is learned by analysing it to improve the product. Through customer feedback, a startup can also validate their

Table 1: Startup components and characteristics (inspired from [22])

Startup component	Characteristic	Description
Institution	Lack of resources	Economical, human, and physical resources are extremely limited
	Small team	Start with a small numbers of individuals
	Low-experienced team	A good part of the development team is formed by people with less than 5 years of experience and often recently graduated students
	New company	The company has been recently created
	Flat organisation	Usually founders-centric and everyone in the company has big responsibilities, with no need of high-management
	Not self-sustained	Especially in the early stage, startups need external funding to sustain their activities
Innovation	Little working history	The basis of an organisational culture is not present initially
	Rapidly evolving	Aim to grow and scale rapidly
	Third party dependency	Due to lack of resources, to build their product, startups heavily rely on external solutions
Extreme uncertainty	One product	Company’s activities gravitate around one product/service only
	Highly reactive	Able to quickly react to changes of the underlying market, technologies, and product
	Time-pressure	Release fast and work under constant pressure
	Highly risky	The failure rate of startups is extremely high

hypotheses. As the result, the startup might pursue a new direction of the business or continue and scale it. Pivot is common to any startup, since it will help the startup from bankruptcy if time between pivots is minimised [24].

- Innovation accounting. To improve the outcomes, entrepreneurs must empirically measure and communicate the real progress of innovation.

Even though Lean startup bears potential to improve the innovativeness of large companies, as Ries argued [24], the scientific and empirical evidences of its application in large software companies are yet to be collected. This study is one of the first efforts to fill this observed knowledge gap.

3. RESEARCH METHODOLOGY

The objective of this study is to explore the application of Lean startup in large software companies context. This study intends to investigate why large software companies use Lean startup to improve their innovativeness. Due to the uniqueness and nascent nature of the research phenomenon, single case study is considered suitable research approach. Since each company is unique, the findings are difficult to generalise and subject to the question of external validity [14].

The case company (Company A for anonymity) is one of the large software companies in Europe. Company A develops antivirus, cloud content and computer security software. It introduced Lean startup in 2012. In the beginning, top management decided to create an internal startup team to enter consumer business with a new idea around people protection. It was an experiment to generate radical innovation since it was a new product for the company itself and was targeted at new market. Three months later, the team released the app to the internal market. Based on the feedback, the team was pivoted to sharing location idea. The new app was released to Google Play, AppStores and Windows Stores. Despite the success of reducing time-to-market, the new app was unable to bring a number of new

users as expected. Moreover, due to the change in the corporate strategy, the new product was already beyond the scope. After a year, the top management stopped the development project and release the software code as open source. However, based on the positive experience obtained, the top management introduced three new internal startups which developed three different products.

We interviewed 8 employees with different roles in the company: 1 internal startup team lead, 1 senior manager, 2 innovation managers, 1 software developer lead, 1 software developer, 1 top manager and 1 UX designer. Each interview lasted between 45 minutes to one hour. All interviews were transcribed verbatim and coded using Corbin and Strauss [31] technique. The main ideas and concepts are identified using open and axial coding. The main components and characteristics of startups discussed in Section 2.1 were used as seed categories to identify the characteristics of internal startups. Meanwhile the Lean startup principles presented in Section 2.3 were used as the analytical framework to make sense of the Lean startup application in the company.

4. RESULTS

4.1 Why internal startup?

As a large company, Company A also does other types of innovation initiatives, for example acquisition. However, acquisition is not an easy task since it takes long time and requires more budget. Acquisition allows the company to increase the number of users but it also comes with negative impact. According to one of the innovation managers, “*It [acquisition] has actually bad things as well. The technology may not be that good. You just buy the name, the app or whatever. For example, if Google bought WhatsApp or Facebook bought WhatsApp, they also buy the infrastructure. The infrastructure may not be that good. Facebook could have done better but that is just the timing issue.*”

On the other hand, creating a new venture is deemed a high risk, as the other innovation manager said, “*If you check just the statistics, 90% fail. If you’re a business guy, you fail there as well. Why would you invest, when you know the like-*

likelihood of failure is 90%?” According to the top manager, the reason to practise Lean startup was to look for growth, by creating new product for new market. It was an experiment because it was the first time for the company to run internal startup. Hence, the investment level was low.

4.2 Characteristics of internal startups

4.2.1 Institution

The internal startup team was formed as a separate and independent team but controlled. The team is separated in the sense that the way of working is different with the rest of the company. The software developer lead said: “... teams that work on their own with ... minimum interaction with the corporate.” However, they are bound by the corporate strategy as the senior manager admitted that the innovation “has absolutely nothing to do with the strategy of the company, there is no point.” However, in the operation, the startup should only focus on the scope defined for them. The top manager said: “[The startup is] in controlled, because the focus is innovation. So in the internal startup, the focus is said, the business case is approved, and they are running towards the target under that business case approval and making experiments, may be even pivot, it's timely kind focus rehearsal.”

Although the company has a long tradition in innovation, working in a startup manner was new to the company at the beginning. Even the top management did not know how to run it. The internal startup team lead said: “He [the top manager] doesn't really know, but [the internal startup team lead] go and find out.” Hence, the team did not get any support for some of their activities. For example, in sales and marketing, the internal startup team lead found that they did not have sufficient budget. Even worse, the sales and marketing department cannot help them, so they had to find a way to do that. However, they got support from other functional entity in the company, e.g. legal, tester, security experts etc.

4.2.2 Innovation

Corporations usually already have a standard operating procedure. Each procedure is performed by a specific person with a specific job description. As an innovation initiative, Lean startup requires new procedures which are different from prior routines. The senior manager said: “As we don't need to follow exactly the same restriction, We don't need to follow the same criteria when it comes to UX, the same hosting standards. So you're more free to do things ...”. However, this raised an internal conflict, as the team lead said, “... because we had to break the rules and that meant we had a lot of people very angry at us”.

Contrary to external startups, internal startups might not be required to use the cutting-edge technology to enter new market. The reason is because as a part of the corporation, they already have customers and markets. The senior manager said: “So, as a corporation, you already have a big business ongoing. And then what you do is to extend that business by adding product, by adding offering, by adding buyer.” They just need to make sure that the new product does not cannibalise the existing products, as she added, “If I go to my manager saying I have the next generation antivirus for example. The first question would be, ok, where is the market? Is it the new market or is it the same one?”

4.2.3 Extreme uncertainty

Internal startups also confront the extreme uncertainty situation, but to a less extent than standalone startups do. Corporation is an ongoing business. The reason it becomes corporation is because it was successful at some stage of its life and it was great at selling and marketing. If the knowledge can be used by internal startups in the creation of new products, they already gained advantages in comparison to external startups.

In addition, the company already has the market share and customers. The senior manager saw this as the advantage for the internal startup. She said: “When you work in a corporation though, I think the failure rate is lower because you already have all the marketing, sales expertise, and because you usually go to the existing customers you already have.”

However, in the case of this particular internal startup in Company A, the team was assigned to develop a new product for a new market segment. Although the customers already know the brand, there is no guarantee that the end customer will pay for it. The senior manager realised that, since the new product was far away from the existing market, it was difficult to incorporate it into their portfolio.

The uncertainties are also recognised by the top manager. They did not have any experience of how to manage an internal startup. He said: “There is no certainty of the business case, just the estimation how we should do, what target segment we are reaching for and ... KPI and metrics.”

4.3 Lean startup principles inside corporation

We identified the lessons learned of practising internal startup inside corporation based on the Lean startup principles described in Section 2.3.

4.3.1 Entrepreneurs are everywhere

It is not difficult to find employees that have big vision about new product or market [23]. One of the innovation managers in Company A found that these employees were passionate about their visions and even willingly work extra hours to prove that, as he said, “Quite often the people end up spending their own time as well because it's something that they are really passionate about.”

The startup initiative was driven by top management. The top management provided a theme for new product development and created a team to implement it. The team explored the feasible concrete idea. The internal startup lead said: “So we were not told to do people tracking service, we were told a theme, a box you need to explore is people protection ... we concluded that the most, the biggest opportunity would reside in this kind eh, small simple location sharing service. And that was what I was pitching to the leadership team in December. They fell in love with the idea because the concept was, it was pretty solid already in that time.”

Like in other innovation initiatives, practising Lean startup inside a large company needs full support from top management. Another innovation manager said: “But, you know, they first need to have full commitment from the board to make something happen. Otherwise it is not experienced because it never gets through. It is very complicated. At least that is my experience in a corporation, it's extremely difficult to have bottom up.”

4.3.2 Entrepreneurship is management

Although the company has R&D, the top management decided to choose internal startup for new product development. The top manager argued that the internal startup has to do some research about the new product and also responsible for its commercialisation. He said: *“Internal startup, yeah, they are doing research as well but not, it’s more execution actually but they are really ... scope in product area and business that they are driving, compared to the research. This is like, you use different [ways] between the [research] results and execution or business execution.”* However, he realised that it was an experiment to work in startup manner.

The internal startup team did not have any obstacle in developing the app. However, it was not the case with the marketing, as the senior manager said, *“So, for example one of the first thing we heard about the marketing manager in my team is like ‘Please don’t release faster than every three months, because I don’t have the visibility.’ ... The faster you can release do it. If you can release every three days do it. But he wanted us to release every three months. So he could understand what’s going on.”*

Moreover, working in a startup manner is not for everybody in the company. Some works better with stability, some works better in uncertainties. The senior manager also said: *“But the problem when we’re recruiting internal people, the number one constraint of everyone was what if it will fail. They all wanted to hear that nothing is gonna happen and we’re gonna be great for years. I mean, the problem is I cannot tell that.”*

4.3.3 Validated learning

The internal startup involved the customers in development phase which reduced the experimentation time to validate the ideas. The team managed to pivot from a specific to common market segment. The senior manager said: *“We created product for hackers so it was very technically oriented product and we have pivoted it because basically what we did was just to change the target. We are targeting basically the mass market now. We don’t want advanced users anymore. What we want are actually the masses.”* Moreover, having close conversation with the customers allowed the team to improve the product, as he added, *“They tell exactly what they need. It’s great. They send us email all the time ... we got 650 emails roughly, which is amazing.”*

To go out of the building needs more budget. As the UX designer argued that to be successful, big and clear vision is not enough. Financial support is also needed to get new customers in. He said: *“In financial part, we haven’t ... it’s all about budget matters. If you have some clear goals lets say, and some kind of budget, you have a chance to success. If not, I do not recommend [internal startup].”*

The other reason for pivoting was not enough end users. One of the lessons learnt was there was no need that would be solved by that product. The top manager said: *“Because the target segment was the kids who are starting school and for the parents that they can see where the kid is. And in [the country] it’s pretty safe. You can walk. You don’t need to drive a car and take the kids to the school. They can walk or by bicycle to school and no harm and there is no accident so far. Really really careless like one or two within ten years. And that’s why there is no such urgent need for [the product] in [the country]. And in Europe maybe more but not enough.”*

In US yes, but in US there are lots of other applications and competitors who solve the same problem, may be better than [the product].”

4.3.4 Build-measure-learn

Working in build-measure-learn loop allowed the team to build the right product. The startup mode increases the speed of development. The senior manager said: *“In our corporation it usually has one product per year with maybe two releases. We have a monthly release minimum. So the speed of development we have is actually six, seven times faster than pretty much any other team in the company.”*

One of the innovation managers said that the term viable in MVP is misleading and suggested that the product should be desirable. He argued that *“Even if it is viable, ... but people just don’t love it from the start, it just doesn’t work.”* Delivering a working software is not enough to compete in a specific market. He said: *“But if you are entering a business where you are competing with Instagram and other fancy application, you have to be fancy at the start. Otherwise, you’ll never know if you lost because of lack of fanciness. So what is minimum? It does not mean you lack of user experience, for example.”*

However, to increase the speed of learning, breaking the rules is inevitable. Following standard architecture and programming practices that potentially slow down the speed of development should be avoided. The internal startup team lead said: *“For example, when we realised that we want to launch in Russia and we have a date, if we follow the ... process we don’t have localisation ready. So then we decided ‘Hey, we will write the texts ourselves.’”*

4.3.5 Innovation accounting

The customers for the internal startup are the end users and top management. Therefore, the team must aim to not only increasing the end users’ perception but also securing the sustainability of the project fund from the management. The internal startup team lead said: *“So basically we were given a time slot. The first time slot was a period of six months and we had to get iPhone and Android versions out and the users’ feedback have to be good ... and the next period was three months and we had to get the Windows phone out”*

Measuring is useful but it is costly. For developers, maintaining several versions of the same app is complicated. For example, regarding the A/B testing, the senior manager said: *“Of course you can do A/B testing, you can do a lot of things but it’s too time consuming and in our case, it was just an experiment. So the likelihood of failure was so high.”*

To track the progress of the startup, the top management always relied on two measures: net promoter score (NPS) and the amount of users. The top manager said that these two measures are enough to decide whether to stop the process or to continue. It is interesting because the reason why they used the measures is because the competitors are also using them even though they develop different types of software. He said: *“We trust NPS and the amount of users. It’s similar a lot with SuperCell and Rovian with AngryBirds fellows, how they are measuring the end user experience and what they are trying, you know, in game, they don’t even release if NPS is low, so they killed it before release.”*

5. DISCUSSION

The case of Company A emerges the differences between

the internal and external startups. Drawing upon Table 1, we summarise these differences in Table 3. From the table, we can see that internal startup depends on corporate strategy. Large companies do not want to harm their existing business and markets. Therefore, the internal startup operates only within the boundary set up by management.

Literature suggests that entrepreneurship will not be successful in large bureaucratic corporation. Innovators and intrapreneurs need an environment where they have autonomy and freedom to innovate without following the existing procedure [19]. Literature (e.g. [15, 17, 19, 27]) recommends to create new venture to bring new products and services to the market and also to build new core competence. The venture can exist internally, as part of the corporation or externally, as a new entity beyond the corporation. Taking into account the four forms of corporate venturing [18], the internal startup in Company A is classified into direct-internal. When the top management decided to create a startup, they also provided the fund and other resources directly to the team. The team members were recruited internally. They had freedom to define the product they wanted to develop as long as it still was within the corporate strategy.

The decision to run corporate venturing is not viewed as a stand-alone activity. In its inception in 2011, the top management identified a potential market segment in people protection. Therefore, an internal startup team was formed to generate the concrete idea within the theme given, as well as implement and commercialise it. Bringing in the venture life-cycle model proposed by [4], the internal startup in company A is in stage 2, which is venture unit. In this stage, the company as a whole does not support entrepreneurship but decides to create a focused and dedicated unit with necessary skills and capabilities to bring new product to market.

Looking back to our research question, our case study shows that applying Lean startup methodology helps the company to build the right product and to find the right market segment faster. Similar to Lean manufacturing, this can be done through waste elimination and continuous flow [24]. In the Lean startup philosophy, waste is everything that beyond what is needed to learn from customers. However, listening too much to existing customers leads to the creation of me-too product which at the end may suppress the radical innovation [8]. Study by Oliver et. al. [21] finds that Japanese products are over-represented in mass market products but under-represented in premium markets. One of the explanation is that the premium product is not driven totally by the customer needs, but by a set of producer's value. In corporation, the corporate strategy plays an important role to keep the new product align with the existing portfolio. It means that the new product still contains the producer's value. Whenever a product has gone beyond the scope, it is better to stop the process or spin it off.

Focus on customer need is the first principle of Lean Thinking as defined by Womack and Jones [32]. However, for internal startups, they have additional types of customers besides the external ones of the product, which is the corporate management. Focusing on the needs of external customers allows the startup team to build the right product while focusing on the needs of corporate management secures the resources needed by the team itself. It seems that the internal startup has two masters to serve. In reality, corporate management plays an important role as the customer as external customers. Therefore, the full commitment from

corporate management is a must for internal startup.

To get the early feedback from customer, Eric Ries suggests to build a MVP [24]. The MVP is a minimum, not a fully complete software. To make it complete, the software must be developed iteratively using the build-measure-learn loop. It is a continuous process to learn what gives value to the customers. In corporation context, our case study shows that implementation of this loop had raised an issue within the company. The internal startup team was getting many complaints from other teams in the company since they did not follow the standard procedure in the company. Therefore, top management support is needed to resolve this situation.

Literature also reveals the importance of championship in every innovation initiative. A champion does not refer to someone on the top management whom always gets the latest report. Rather it means someone in a senior position who can protect the initiative whenever problems arise, including the change in high-level strategic [1, 4]. In our case study, the presence of the champion is not recognised. This was proven by the absence of top management protection to the internal startup when the corporate strategy was changed. Eventually, the startup was terminated and the product was released as an open source software.

6. CONCLUSIONS AND FUTURE WORKS

In this study, we reported on a study that investigates the experience of a large software company using Lean startup to improve their innovativeness. A single case study was conducted in which we interviewed 8 employees with different roles. The study examines the characteristics of internal startups and the implementation of Lean startup principles in large software companies.

This paper makes several contributions. First, it examines the differences between internal startups and external startups. The second contribution is that, using the Lean startup principles as an analytical framework, we identified the lessons learned on practising Lean startup in large companies.

We envision a future study which will investigate in-depth Lean startup in practise. Moreover, a comparative study on the companies that have been successful with Lean startup is put into the next agenda.

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Table 3: Differences in characteristics between external and internal startups

Component	External startup	Internal startup
Institution	Lack of resources, small team, low-experienced team, new company, flat organisation, not-self sustained, little working history	High-experienced team, controlled (operates in the approved business case), full of resources (including monthly salary)
Innovation	Rapidly evolving, third-party dependency, one product	Targeting on new market segment (fear of cannibalising)
Extreme uncertainty	Highly reactive, time-pressure, highly risk	Breaking the law, highly depends on the corporate strategy (controlled)

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